

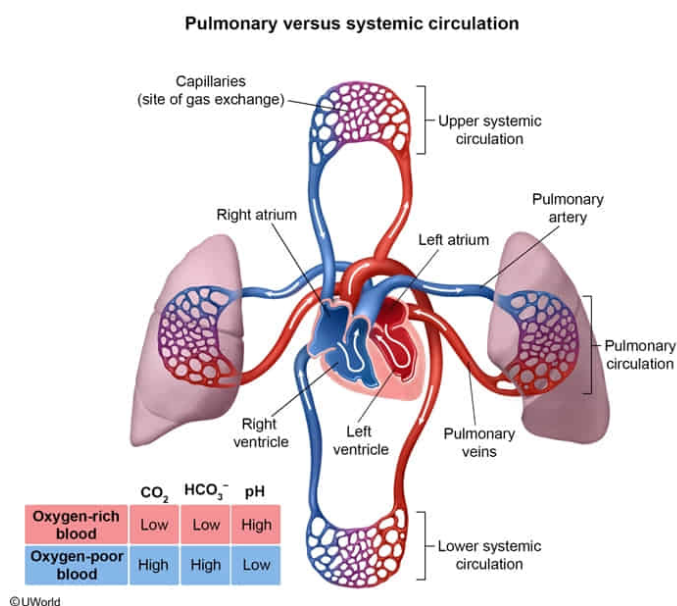
Which of the following is generally true about the path of blood flow through the heart?

- ☐ A. The right ventricle contracts to propel blood to all body tissues except the lungs. [4%]
- ☐ B. The left ventricle pumps blood into the left atrium. [8%]
- ✓ ☒ C. The aorta receives blood from the left ventricle. [76%]
- ☐ D. The right atrium receives blood from the pulmonary veins. [10%]

Correct

76% answered correctly

Explanation:



The heart is a muscular structure that uses strong contractions to propel blood continuously throughout the circulatory system. The heart is composed of two separate pumps that contract simultaneously; the right side of the heart pumps blood to the lungs, and the left side of the heart pumps blood to the rest of the body. The heart has four chambers: The left and right atria are receiving chambers, whereas the left and right ventricles are pumping chambers that supply the force necessary to propel blood through the circulatory system. The atrioventricular (AV) and semilunar (SL) **valves** prevent the backward flow of blood into the atria and ventricles, respectively. Accordingly, blood flows through the circulatory system as follows:

1. Deoxygenated blood from body tissues enters the right atrium via the (superior and inferior) vena cavae (**Choice D**).
2. The right atrium contracts, forcing the *deoxygenated* blood through the tricuspid valve and into the right ventricle.
3. The right ventricle contracts to propel the blood through the pulmonary valve and into the pulmonary arteries (**Choice A**).
4. The pulmonary arteries carry blood from the heart to the capillary beds of lung alveoli, where gas exchange occurs.
5. In pulmonary capillary beds, the blood unloads carbon dioxide and loads oxygen (ie, becomes oxygenated) and returns to the *left atrium* via the pulmonary veins.
6. The left atrium contracts, pushing the oxygenated blood through the bicuspid (mitral) valve and into the left ventricle (**Choice B**).
7. The left ventricle contracts to pump blood through the aortic valve to the rest of the body via the aorta.
8. The blood flowing through systemic blood vessels delivers nutrients and oxygen to the body tissues before returning to the heart.

Educational objective:

The right heart receives deoxygenated blood from the body and pumps it out toward the lungs. The left heart receives oxygenated blood from the lungs and pumps it out toward the rest of the body. Atria receive blood; ventricles release blood.

Time spent:0 sec

Subject:
Biology

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System:
Circulation and Respiration